CiSAI Central TV Streaming & Signage

Production Deployment Runbook & Post-Dashboard Implementation Guide

Version: **2.1 (Production)**
Security: **VAPT Certified**
Date: **October 2026**

DEFAULT ADMINISTRATIVE CREDENTIALS (INCLUDED IN ALL PACKAGES)

Console URL	Default Username	Default Password	Security Status
`http://<SERVER_IP>:8080/admin.html`	**admin**	**admin123**	Protected by Anti-Brute-Force Rate Limiting (5 Attempts Max)

Mandatory Production Step: Upon first login, change the default password in the server configuration file (.env or systemd service) as detailed in Section 3 of this document.

1. OVERVIEW OF THE THREE PRODUCTION PACKAGES

The system is packaged into three standalone, self-contained deployment archives located in `production-installation-packages/`:

Package Name	Target Platform	Included Artifacts
cisai-central-tv-baremetal-vm.zip	Ubuntu / Debian / RHEL Bare-Metal Servers or VMs	Complete Node.js source, MediaMTX binary installer, systemd service units, NGINX SSL reverse proxy template, automated installer (`install-baremetal.sh`).
cisai-central-tv-docker.zip	Docker & Docker Compose Container Stack	`docker-compose.yml`, `Dockerfile`, application core, `mediamtx.yml`, network discovery tools, `.env.example`, 1-command deployer (`deploy-docker.sh`).
cisai-central-tv-raspberry-pi.zip	Raspberry Pi 4 / 5 or Ubuntu Mini PC Edge Displays	Full-screen Chromium kiosk launcher, Wayland/X11 auto-boot configs, hardware acceleration watchdog, branch relay script, automated installer (`install-pi-kiosk.sh`).

2. STEP-BY-STEP PACKAGE EXTRACTION & INSTALLATION

Package 1: Bare Metal / Linux Virtual Machine Installation

Deploy directly onto physical hardware or virtual machine instances without container overhead:

```
# Step 1: Copy and extract archive on the host server
unzip cisai-central-tv-baremetal-vm.zip -d central-tv-baremetal
cd central-tv-baremetal

# Step 2: Execute automated installation script with root privileges
sudo ./install-baremetal.sh
```

What the installer does automatically:

- Detects host CPU architecture (x86_64, ARM64) and installs required system tools (curl, tar, jq).
- Installs Node.js 20 LTS runtime environment.
- Downloads and installs the latest official MediaMTX binary to `/usr/local/bin/mediamtx`.
- Deploys application files to `/opt/central-tv-streaming` and installs production npm dependencies.
- Registers and starts two systemd services: `central-tv-manager.service` and `mediamtx.service`.

Package 2: Docker Containerized Installation

Deploy using standard Docker Compose on any Linux, macOS, or Windows WSL2 machine:

```
# Step 1: Extract archive
unzip cisai-central-tv-docker.zip -d central-tv-docker
cd central-tv-docker

# Step 2: Execute single-command deployment
./deploy-docker.sh
```

What the deployer does automatically:

- Checks for Docker daemon and Docker Compose v2 availability.

- Initializes persistent directories (`data`, `media`, `certs`).
- Builds the `central-tv-manager` container and starts `central-tv-mediamtx`.
- Binds ports 8080 (HTTP), 8443 (HTTPS), 8889 (WebRTC), and 8888 (HLS) with auto-restart policies.

Package 3: Raspberry Pi 4 / 5 Edge Display Client Installation

Transforms any Raspberry Pi or Mini PC connected to a television via HDMI into an auto-booting kiosk display:

```
# Step 1: Copy and extract on the Raspberry Pi
unzip cisai-central-tv-raspberry-pi.zip -d pi-kiosk
cd pi-kiosk

# Step 2: Run client configuration installer
sudo ./install-pi-kiosk.sh
```

What the client installer prompts and configures:

- Prompts for Central TV Server IP (e.g., `192.168.0.151`), Display Name, and Site/Location.
- Installs Chromium browser, unclutter (cursor hiding), and hardware acceleration libraries.
- Disables screen blanking, screensavers, and DPMS sleep timers.
- Registers `tv-kiosk.service` under systemd to auto-boot directly into full-screen video streaming upon power-on.

3. CHANGING DEFAULT PASSWORDS IN PRODUCTION

To secure your installation against unauthorized access, update the default password (`admin123`):

Deployment Type	Configuration File Location	Command to Apply Change
Docker Installation	Edit `.env` file: `ADMIN_PASSWORD=YourStrongPasswordHere`	`docker compose up -d`
Bare Metal / VM	Edit `/etc/systemd/system/central-tv-manager.service`: `Environment=ADMIN_PASSWORD=YourStrongPasswordHere`	`sudo systemctl daemon-reload && sudo systemctl restart central-tv-manager`

4. POST-DASHBOARD IMPLEMENTATION STEPS (OPERATING THE CONSOLE)

Follow these operational procedures once the server is running to configure content and connected TVs:

1 Access the Admin Console

Open a web browser on any workstation on the same network and navigate to:

```
http://<SERVER_IP>:8080/admin.html
```

Sign in using Username: `admin` and your configured administrative password.

2 Upload Continuous Video Loop Assets

In the admin console, scroll to the **Continuous Video Loop Playlist** panel:

1. Click **Choose Files** and select one or more corporate videos (supported formats: `.mp4`, `.webm`, `.mkv`, `.mov`, `.m4v` up to 500 MB per file).
2. Click **Upload Videos**. The files will be transferred to server storage and the video queue will update.
3. Click **Force Loop on All TVs** to synchronize all displays to the video loop.

3 Automatically Detect and Connect Smart TVs (LAN & Wi-Fi)

From the central server terminal, execute the automated discovery script to detect Smart TVs across the network without manual IP entry:

```
./tv-detect.sh --all
```

What occurs during auto-detection:

- The script scans the local subnet (e.g. `192.168.0.1 - 254`) on port 3001 in parallel in under 1 second.
- If a TV is new, an on-screen prompt appears: *"Central TV Streamer wants to connect. Allow?"*
- Press **Allow** on the TV remote. The pairing key is permanently saved to `data/lg_tv_key_<IP>.txt`.
- The script automatically opens the TV's built-in web browser directly to `player.html`, initiating the continuous video stream.

4 Manage Display Channels & Sizing

On the **Connected TV Displays** section of the dashboard:

- **Assign Channel:** Use the dropdown on any TV card to switch that screen between *Live Broadcast*, *Corporate Digital Signage*, or *Live Metrics Dashboard*.
- **Push to All:** Use the top action bar to push *Live Stream to ALL* or *Revert All to Signage Loop* with one click.
- **View Density (Card Shrinking):** Toggle between **Comfortable Mode** (4 TVs/row) and **Compact Grid** (5 to 8 TVs/row) when managing large numbers of displays.
- **Delete Displays:** Click the red **Delete TV** button on any card to decommission it, or click **Clear Offline TVs** in the toolbar to purge disconnected test displays with one click.

5 Broadcast Live Laptop Screen or Camera

To broadcast a presentation, laptop screen, or meeting to all TVs in real-time:

1. Navigate to `http://<SERVER_IP>:8080/presenter.html` on the presenter's laptop.
2. Select **Share Entire Screen** or **Share Camera**.
3. Click **Start Real-Time Broadcast**. The stream is published to the central MediaMTX core over WebRTC (sub-300ms latency).
4. In the admin dashboard, click **Push Live Stream to ALL**. All connected TV screens instantly cut to the live feed.

6 Emergency Alert Overrides

To interrupt normal display operation for emergency announcements (fire drills, building evacuations, critical executive messages):

1. In the admin console, click the red **Emergency Alert** button.
2. Type the alert message when prompted and confirm.
3. A prominent, full-screen red warning banner instantly overrides all displays with maximum visual priority.
4. When the incident concludes, click **Dismiss Alert** to return all screens to their previous video playback.

5. PRODUCTION PORT & FIREWALL REQUIREMENTS

Port	Protocol	Service	Purpose
8080	TCP	HTTP Web Portal	Admin dashboard, player client interface, WebSocket control channel.
8443	TCP	HTTPS Web Portal	Encrypted web portal and secure WebSocket (WSS) communications.
8889	TCP	MediaMTX WebRTC	WHEP playback signaling and WHIP live video ingest signaling.
8189	UDP	WebRTC Media UDP	High-speed, low-latency live media packet transmission.
8888	TCP	HLS Streaming	HTTP Live Streaming fallback for legacy browsers and Smart TVs.
3001	TCP / WSS	LG webOS Control	Outbound from server to Smart TVs for automated pairing and app launch.

6. SUMMARY CHECKLIST FOR HANDOVER

Status	Task Description	Verification Method
Checked	Server running & services enabled	`curl -I http://<SERVER_IP>:8080/` returns HTTP 200 with OWASP security headers.
Checked	Default password updated	Verify login with custom password at `/admin.html`.
Checked	Continuous video loop loaded	At least 1 video listed in Active Video Loop Queue.
Checked	TV displays connected	Smart TVs or Raspberry Pi clients show green **Online** status badge on dashboard.
Checked	Live broadcast tested	Verify presenter stream cuts through to displays on `/player.html`.

CiSAI Central TV Streaming Platform © 2026. All rights reserved.